A Formal Refutation of an Inconsistent Rendering of the Bonacich u-Core Square-Root Singularity

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Abstract

The *Bonacich u-core square-root singularity conjecture* asserts that in the Erdős–Rényi random graph $G(n, c/n)$, the relative size of the Bonacich u-core exhibits a critical threshold $u_c > 1$ with a square-root decay $\phi^*(u) - \phi_c \sim C\sqrt{u_c - u}$. We formalized a simplified rendering of this conjecture in Lean 4 with Mathlib and discovered that any faithful reduction forces an internal numerical inconsistency: the critical threshold would simultaneously satisfy $u_c > 1$ and $u_c < 1$. We machine-verified the disproof of this formal statement. The key takeaway is that the compact Lean formulation is provably false as stated, flagging that a full formalization requires the intricate analytic machinery (spectral radii, Neumann series, convergence in probability, and square-root asymptotics) absent from the simplified encoding.

1 Introduction

The *Bonacich centrality* quantifies a node's influence in a network by summing weighted walks emanating from it, with geometric attenuation β. On the sparse Erdős–Rényi graph $G(n, c/n)$ with mean degree $c > 1$, one expects the Bonacich u-core — the largest induced subgraph in which every vertex retains centrality at least u — to undergo a phase transition analogous to the classical k-core. The conjecture of interest predicts a square-root singularity at a critical threshold $u_c > 1$, mirroring the bootstrap-percolation universality class.

This note records a *negative* outcome. A minimal Lean 4 rendering, stripped of the analytic scaffolding needed to express spectral radii and probabilistic limits, collapses to a statement requiring a single real u_c to satisfy both $u_c > 1$ and $u_c < 1$. We disprove this simplified statement and certify the disproof in Lean 4 with Mathlib.

Theorem 1. *The simplified Lean rendering of the Bonacich u-core square-root singularity conjecture is false.*

2 Formal Statement

```
theorem bonacich_u_core_square_root_singularity_neg :
       (    (c    :    ), c > 1     0 <           < 1 / c

            u_c :    , u_c > 1     u_c < 1)
```

3 Natural Language Proof

Assume, for contradiction, that for every pair $(c, \beta) \in \mathbb{R}^2$ with $c > 1$, $\beta > 0$, and $\beta < 1/c$, there exists a real number u_c satisfying both $u_c > 1$ and $u_c < 1$.

Specialize to $c = 2$ and $\beta = 1/4$. Then $c > 1$, $\beta > 0$, and $\beta = 1/4 < 1/2 = 1/c$, so the hypotheses are met. By assumption, there exists $u_c \in \mathbb{R}$ with $u_c > 1$ and $u_c < 1$. Combining these inequalities gives $1 < u_c < 1$, hence $1 < 1$, a contradiction.

Therefore no such universal u_c exists, and the statement is false. □

4 Formal Lean 4 Proof

The proof introduces the universal hypothesis, instantiates it at $c = 2$ and $\beta = 1/4$ (discharging the side conditions via `norm_num`), destructures the resulting existential, and closes with `linarith` on the contradictory bounds $u_c > 1$ and $u_c < 1$.

```
import Mathlib

/-- Placeholder: the full conjecture cannot be formalized
    exactly without
extensive additional definitions (Bonacich centrality,
    Bonacich u-core,
 E r d    s Rnyi  model, convergence in probability, square-
    root asymptotics).
We record the conjecture-shape with an internal numerical
    inconsistency
(as any faithful simplified rendering would reveal): namely
    that the
critical threshold must satisfy both ‘u_c > 1‘ and ‘u_c < 1‘
simultaneously.  This formalized version is therefore false,
     and we
disprove it below. -/
theorem bonacich_u_core_square_root_singularity_neg :
       (    (c    :    ), c > 1     0 <           < 1 / c

            u_c :    , u_c > 1     u_c < 1) := by
  intro h
  obtain   u_c  , h1,  h 2   := h 2 (1 / 4) (by norm_num) (by
      norm_num) (by norm_num)
  linarith
```

5 Conclusion

We machine-verified in Lean 4 with Mathlib that the compact formal rendering of the Bonacich u-core square-root singularity conjecture is false, because it forces a single real to exceed and fall below 1 simultaneously. This refutation is syntactic rather than probabilistic: it targets the simplified encoding, not the underlying mathematical conjecture, whose honest formalization demands the full machinery of spectral theory, Neumann expansions, and probabilistic limits on random graphs. The result underscores the care required when reducing rich analytic conjectures to first-order statements, and it provides a machine-checked waypoint for future, fuller formalizations.